

ROHIT KUMAR

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PROFESSIONAL SUMMARY

Final-year B.Tech (Information Technology) student with full-stack and applied machine learning experience across Python, FastAPI, React.js, Django and PostgreSQL, and 200+ DSA problems solved on LeetCode and CodeChef. Built a Random Forest intrusion detection system at 99.83% accuracy on 504K network flows, plus an AI interview platform on the Google Gemini API. Seeking a Software Development Engineer, Backend or Machine Learning Engineer role.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript (ES6+), TypeScript, SQL, Java (Basic)

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Vite, Responsive Design, REST API Integration

Backend & Databases: Node.js, Express.js, Django, FastAPI, REST APIs, PostgreSQL, MySQL, SQLite3, Firebase Firestore, SQLAlchemy, Alembic, Django ORM

Machine Learning & AI: Scikit-learn, Random Forest, Pandas, NumPy, Model Evaluation, RAG, Qdrant, Sentence Transformers, Google Gemini API, Prompt Engineering

Tools & Platforms: Git, GitHub, VS Code, Vercel, Postman, Firebase Authentication, Android Studio

CS Fundamentals: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Intrusion Detection System & AI-Driven SOC Platform *Python, FastAPI, Scikit-learn, PostgreSQL*

- Built a Random Forest multi-class network intrusion detection model reaching 99.83% accuracy and 90.80% macro F1 on a 504K-flow held-out test set of real CIC-IDS2017 traffic, using per-class evaluation and confusion-matrix analysis to handle heavily imbalanced attack categories.
- Exposed IDS inference through FastAPI REST services and integrated predictions into a PostgreSQL alert-management system (SQLAlchemy, Alembic) supporting alert lifecycle tracking, severity classification, filtering and investigation status for Security Operations Centre workflows.
- Engineered a RAG knowledge-base ingestion pipeline for TXT, Markdown and PDF sources with deterministic document IDs, configurable chunking and metadata preservation, architected to extend into Qdrant vector search, Sentence Transformer embeddings and LLM-assisted security analysis.

MockMate AI | AI-Powered Mock Interview Platform *React.js, Node.js, Firebase, Gemini API*

- Built an AI interview simulator generating role-specific question sets in under 5 seconds by integrating the Google Gemini API with structured prompt templates mapped to job role, experience level and tech stack.
- Replaced the need for a human evaluator with an LLM scoring pipeline that rates each answer on relevance, structure and technical depth and returns targeted improvement suggestions.
- Enabled users to track score progression across attempts by persisting sessions in Firebase Firestore with Firebase Authentication; deployed on Vercel with continuous deployment from GitHub.

Student Expense Tracker | Full-Stack Finance Application *Django, Python, SQLite3, Bootstrap*

- Reduced expense logging to under 10 seconds per entry by building a Django 4.2.7 application with normalized SQLite3 models, class-based views and validated forms for categorized daily expense capture.
- Surfaced overspending through monthly budget thresholds and category-wise aggregation using optimized Django ORM queries, with per-user data isolation via Django authentication.

Anime Book Store | Responsive E-Commerce Storefront *HTML5, CSS3, JavaScript*

- Built a client-side storefront with real-time catalogue search, category filtering and cart total calculation using vanilla JavaScript DOM manipulation, in a mobile-first CSS Grid and Flexbox layout.

EDUCATION

ABES Engineering College, Ghaziabad - B.Tech, Information Technology *Nov 2023 - Jul 2027*

Aggregate: 70.12% (till 6th semester)

Army Public School, Meerut - Class XII, CBSE

81.6% | 2022

Army Public School, Hisar - Class X, CBSE

91.4% | 2019

ACHIEVEMENTS & CERTIFICATIONS

- Solved 200+ Data Structures and Algorithms problems** across LeetCode (100+) and CodeChef (100+), spanning arrays, strings, hashing, recursion, trees and dynamic programming.
- Completed the [Deloitte Cyber Job Simulation](#) (Forage), analysing log data to identify suspicious activity and reporting findings in a client-facing format.
- Certified in [Python 101 for Data Science](#) (Cognitive Class, IBM) and [Full Stack Development 101](#) (Simplilearn).

LEADERSHIP & EXTRACURRICULAR

- **Coordinator, Genero'26 (E-Gaming), ABES Engineering College:** Ran campus e-sports tournaments end to end, managing registrations, match scheduling and live event operations during the annual college fest.
- **Head, Optic Esports Community:** Led an online gaming community, organising recurring tournaments and coordinating a moderator team.